

**GOSFORD HIGH SCHOOL
YEAR 12 - 2000**

HALF YEARLY EXAMINATION

MATHEMATICS

***3 Unit (Additional)
&
3/4 Unit (Common)***

Time Allowed - 2 hours
(plus 5 mins .reading time)

Directions to Students:

- * Attempt **ALL** questions
- * All questions are of equal value
- * All necessary working is to be shown, in every question. (Marks may be deducted for careless or badly arranged work)
- * Board approved calculators may be used
- * Standard integral sheets are supplied
- * Start each question on a **SEPARATE** page

Question 1

- a) Differentiate $2x \tan^{-1} x$
- b) Find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x dx$
- c) Find the acute angle between the lines $3y = 2x + 8$ and $y = 5x - 9$.
- d) If α, β, γ are the roots of the polynomial $2x^3 - 14x - 1 = 0$, find $\alpha\beta\gamma$
- e) Prove the identity $\frac{2 \tan A}{1 + \tan^2 A} = \sin 2A$
- f) Evaluate $\int_2^{10} \frac{x}{\sqrt{x-1}} dx$ using the substitution $x = t^2 + 1$

Question 2

- a) i) By considering the sum of the terms of an arithmetic series show that
- $$(1 + 2 + 3 + \dots + n)^2 = \frac{1}{4} n^2 (n+1)^2$$
- ~~ii)~~ ii) By using the Principle of Mathematical induction prove that $1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$ for all $n \geq 1$.
- b) i) The polynomial equation $P(x) = 0$ has a double root at $x = a$. By writing $P(x) = (x - a)^2 Q(x)$, where $Q(x)$ is a polynomial, show that $P'(a) = 0$
- ii) Hence or otherwise, find the values of a and b if $x = 1$ is a double root of $x^4 + ax^3 + bx^2 - 5x + 1 = 0$

Question 3

- a) Sketch a curve which has all of the following properties:

$$\begin{array}{llll}
 f(-4) = 0 & f(8) = 0 & f'(5) > 0 & f'(-3) < 0 \\
 f'(5) = 0 & f''(-6) < 0 & f''(-1) > 0 & f''(0) = 0 \\
 \text{and} & f''(5) < 0 & &
 \end{array}$$

- b) Solve $\frac{1}{1-x^2} \leq 4$ and graph the solution set on the number line.
- c) i) Find the co ordinates of the point of intersection of the curves $y = \sin^{-1} x$ and $y = \cos^{-1} x$.
- ii) Find the angle between the tangents to these curves at their point of intersection.

Question 4

- a) Find the co-ordinates of the point which divides the line joining the points $(-1, 3)$ and $(5, -7)$ externally in the ratio 4:3.
- b) Sketch the graph $|x| + |2y| = 8$
- c) P and Q are points on the parabola $x^2 = 4ay$ with parameters p and q . PQ subtends a right angle at the origin.
- i) Show that $pq + 4 = 0$.
- ii) Find the co-ordinates of M , the mid-point of PQ .
- iii) Find the locus of M .

Question 5

- a) Sketch the function $\sin^{-1} 2|x|$ for $-\frac{1}{2} \leq x \leq \frac{1}{2}$

- b) i) Write down the range of the function $y = \frac{-1}{2(1+x^2)}$
- ii) Write down the range of $\sin^{-1} \left[\frac{-1}{2(1+x^2)} \right]$
- c) $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$. the line L is a tangent at P. S is the focus. Show that L is equally inclined to SP and the axis of the parabola.
- d) Use mathematical induction to prove that, for every positive integer n, $13 \times 6^n + 2$ is divisible by 5.

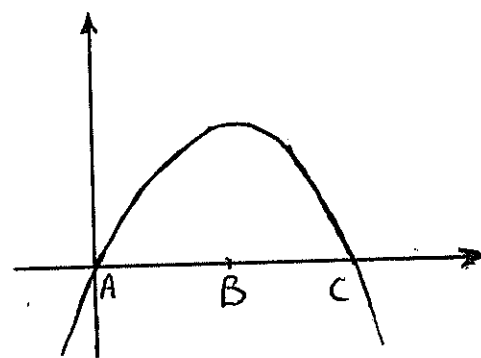
Question 6

- a) Find $\int \frac{dx}{\sqrt{1-4x^2}}$, using the substitution $x = \frac{1}{2} \sin \Theta$
- b) A piece of wire 24 cm long is cut into two pieces. The first is bent to form a square of side x cm and the second is bent to form a circle.
- i) Write down an expression for the sum of their areas.
- ii) Find the greatest value for the sum of the areas.

c) Solve $\frac{1}{|3-x|} \geq \frac{1}{2}$

- d) The sketch shows $y = f^1(x)$

Sketch i) $y = f(x)$
 ii) $y = f^1(x)$
 and iii) $y = f''(x)$



down the page, showing clearly the shape and position of each graph at the points which correspond with A, B and C in the graph of $y = f^1(x)$ shown.

Question 7

- a) The roots of $x^2 + mx + n = 0$ are α, β
Find the values, in terms of m and n of
- $\alpha + \beta$
 - $\alpha \beta$
 - $\alpha^2 + \beta^2$
 - $3\alpha^2 \beta + 3\alpha \beta^2$
- b) Ascertain whether the following statements are TRUE or FALSE.
Use diagrams and words of the English language to support your answers.
- $\int_{-1}^1 x^2 \cos x dx = 2 \int_0^1 x^2 \cos x dx$
 - $\int_{-\pi}^{\pi} (x \cos x + \sin x) dx = 0$
- c) Differentiate $2x \tan^{-1} x - \ln(1+x^2)$ and hence find $\int_0^1 \tan^{-1} x dx$
- d) Consider the circle $x^2 + y^2 - 2x - 14y + 25 = 0$
- Show that if the line $y = mx$ intersects the circle in two distinct points, then $(1+7m)^2 - 25(1+m^2) > 0$.
 - For what values of m is a line $y = mx$ a tangent to the circle?

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}).$$

Note: $\ln x = \log_e x$, $x > 0$

1,2 Rick Q3,4 Tony Q5,6 Chris Q7 Allan

Year 12 Half Yearly - 3 Unit Maths (Solutions)

Question 1 (12 mks)

1) (a) $\frac{d}{dx} (2x \tan^{-1} x)$
 $= \tan^{-1} x (2) + 2x \cdot \frac{1}{1+x^2}$
 $= 2 \tan^{-1} x + \frac{2x}{1+x^2}$

(2) (b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x \, dx$
 $= [\tan x]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$
 $= \tan \frac{\pi}{3} - \tan \frac{\pi}{6}$
 $= \sqrt{3} - \frac{1}{\sqrt{3}}$

(2) (c) $3y = 2x + 8 \rightarrow y = \frac{2}{3}x + \frac{8}{3}$
 $m_1 = \frac{2}{3}$

$y = 5x - 9 \rightarrow m_2 = 5$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{5 - \frac{2}{3}}{1 + 5 \times \frac{2}{3}} \right| = \frac{4\frac{1}{3}}{4\frac{1}{3}}$

$= 1$

$\therefore$ Acute angle between lines = 45°

(1) (d) $2x^3 - 14x - 1 = 0$

$\alpha/\beta = -\frac{d}{a} = -\frac{-1}{2}$
 $= \frac{1}{2}$

(2) (e) Prove that $\frac{2 \tan A}{1 + \tan^2 A} = \sin 2A$

LHS = $\frac{2 \tan A}{\sec^2 A} = 2 \frac{\sin A}{\cos A} \times \frac{1}{\sec^2 A}$

$= 2 \frac{\sin A}{\cos A} \cdot \cos^2 A$

$= 2 \sin A \cos A$

$= \underline{\underline{\sin 2A}}$

(3) (f) $\int_2^{10} \frac{x}{\sqrt{x-1}} \, dx = I$

Put $x = t^2 + 1$ } When $x=2$
 $\frac{dx}{dt} = 2t$ } $t=1$

$I = \int_1^3 \frac{t^2+1}{\sqrt{t^2+1-1}} \frac{dx}{dt} dt$ } When $x=10$
 $t=3$

$= \int_1^3 \frac{t^2+1}{t} \cdot 2t \, dt$

$= \int_1^3 (2t^2 + 2) \, dt$

$= \left[\frac{2t^3}{3} + 2t \right]_1^3$

$= F(3) - F(1)$

$= (2 \times 9 + 6) - \left(\frac{2}{3} + 2 \right)$

$= 24 - 2\frac{2}{3}$

$= \underline{\underline{21\frac{1}{3}}}$

Question 2 (12 mks)

(6) (a) i) $(1 + 2 + 3 + \dots + n)^2$

$= (\text{Sum of an A.P.})^2$

$= \left[\frac{n}{2} (a + l) \right]^2$

$= \left[\frac{n}{2} (1 + n) \right]^2$

$= \underline{\underline{\frac{n^2}{4} (1 + n)^2}}$

(ii) Let $S_n = 1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2$

Step 1: Show that S_n is true for $n=1$

LHS = $1^3 = 1$ RHS = $(1)^2 = 1$

S_n is true for $n=1$

Step 2: Assume S_n true for $n=k$

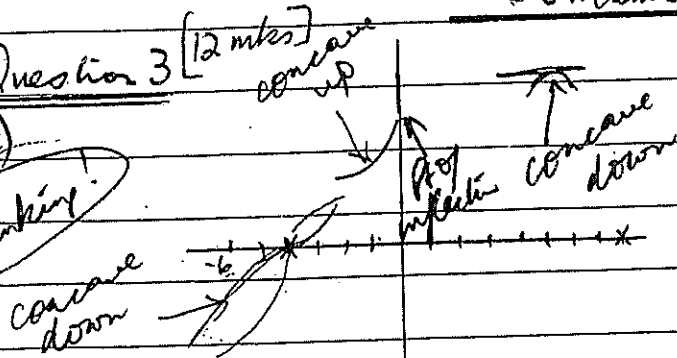
i.e. that

$1^3 + 2^3 + \dots + k^3 = (1 + 2 + \dots + k)^2$

30 Maths Solns (Page 3)

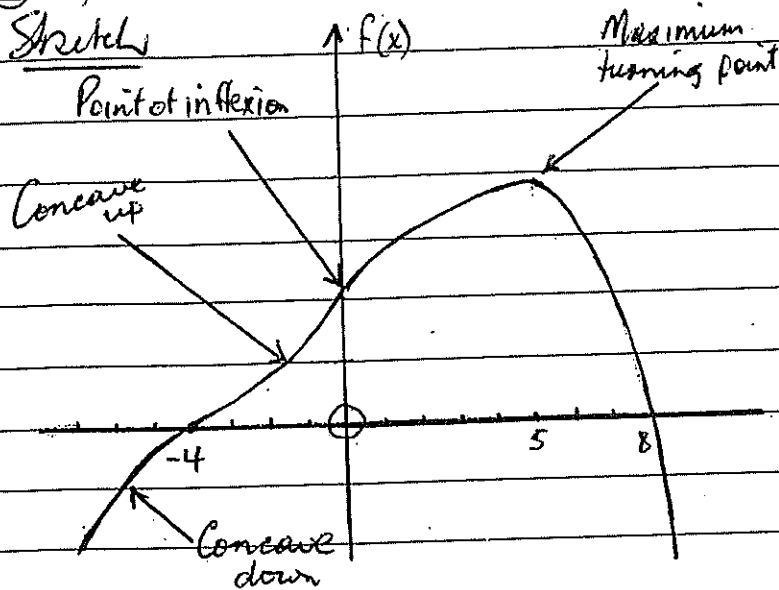
Question 3 [12 mks]

a) Thinking!



(4) (a)

Sketch



(b) (4)

$$\frac{1}{1-x^2} \leq 4$$

$$1-x^2 \geq \frac{1}{4}$$

Solve $1-x^2 = \frac{1}{4}$

$$x^2 = \frac{3}{4}$$

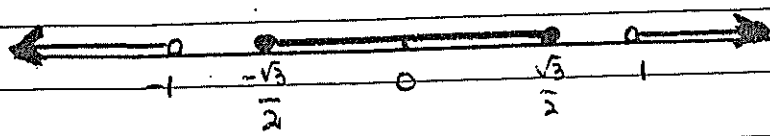
$$x = \pm \frac{\sqrt{3}}{2}$$

Also $1-x^2 < 0$ (cannot equal 0)

$$\text{ie } x^2 > 1$$

$$x > 1 \text{ and } x < -1$$

[Plot the four points on the number line and test regions]

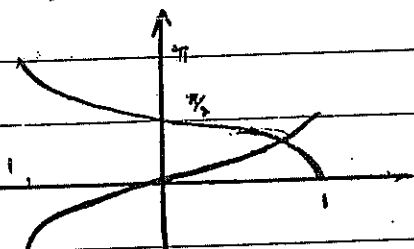


Solution is

$$x < -1, -\frac{\sqrt{3}}{2} \leq x \leq \frac{\sqrt{3}}{2}, x > 1$$

(c) i)

(4) $\times \frac{2}{2}$



$$\frac{\pi}{4} = \sin^{-1} \frac{1}{\sqrt{2}}$$

$$\frac{\pi}{4} = \cos^{-1} \frac{1}{\sqrt{2}}$$

Point of intersection is

$$\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4} \right)$$

OR solve $x = \sin y$ $x = \cos y$ simultaneously.

ii) $y = \sin^{-1} x$
 $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

$y = \cos^{-1} x$
 $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

$$= \frac{1}{\sqrt{1-\frac{1}{2}}} \text{ at } x = \frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{\frac{1}{2}}} = \frac{1}{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{2}$$

$$\therefore m_1 = \sqrt{2} \quad m_2 = -\sqrt{2}$$

Angle between the tangents to the curves at $x = \frac{1}{\sqrt{2}}$

given by

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\sqrt{2} - (-\sqrt{2})}{1 - 2} \right|$$

$$= \left| \frac{\sqrt{2} + \sqrt{2}}{-1} \right|$$

$$= 2\sqrt{2}$$

$$\theta = \tan^{-1} 2\sqrt{2}$$

$$= \underline{\underline{70^\circ 32'}}$$

30 Maths Solution [Page 2]

Step 3 Prove S_n true for $n=k+1$ i.e. $P'(a) = 0$

ie that

$$1^3 + 2^3 + \dots + k^3 + (k+1)^3 \\ = [1+2+\dots+k+(k+1)]^2$$

$$LHS = (1+2+3+\dots+k)^2 + (k+1)^3$$

using assumption Step 2

$$= \frac{1}{4} k^2 (k+1)^2 + (k+1)^3$$

using part (i)

$$= \frac{1}{4} (k+1)^2 [k^2 + 4(k+1)]$$

$$= \frac{1}{4} (k+1)^2 [k^2 + 4k + 4]$$

$$= \frac{1}{4} (k+1)^2 (k+2)^2$$

$$RHS = [1+2+\dots+k+(k+1)]^2$$

$$= \frac{(k+1)^2 (k+1+1)^2}{4}$$

using part (i)

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

$\therefore S_n$ is true for $n = k+1$

Step 4 Since S_n is true for $n=1$ (Step 1)

then from steps 2 and 3

it is also true for $n=1+1$

ie $n=2$ and so on

for all $n \geq 1$

(b) i) $P(x) = (x-a)^2 Q(x)$

⑥ $\frac{2}{4}$ $P'(x) = v u' + u v'$

$$= Q(x) \cdot 2(x-a)^1 \cdot 1$$

$$+ (x-a)^2 Q'(x)$$

$$= (x-a)[2Q(x) + (x-a)Q'(x)]$$

$$\therefore P'(a) = (a-a)[\dots] = 0$$

(ii) $x^4 + ax^3 + bx^2 - 5x + 1 = 0$

$$P(x) = 0$$

$$P'(x) = 4x^3 + 3ax^2 + 2bx - 5$$

If $x=1$ is a double root

$$\text{then } P'(1) = 0$$

$$\text{ie } 4 + 3a + 2b - 5 = 0$$

$$3a + 2b = 1 \quad (1)$$

Also, since $x=1$ is a root

$$P(1) = 0$$

$$\text{ie } 1 + a + b - 5 + 1 = 0$$

$$a + b = 3 \quad (2)$$

Solving simultaneously

$$3a + 2b = 1 \quad (1)$$

$$3a + 3b = 9 \quad (3)$$

$$(3) - (1); \quad b = 8$$

$$\text{and } a = 3 - b$$

$$= 3 - 8$$

$$= -5$$

$$\text{ie } \underline{\underline{a = -5, b = 8}}$$

30 Math Solutions (Page 4)

Question 4 [12]

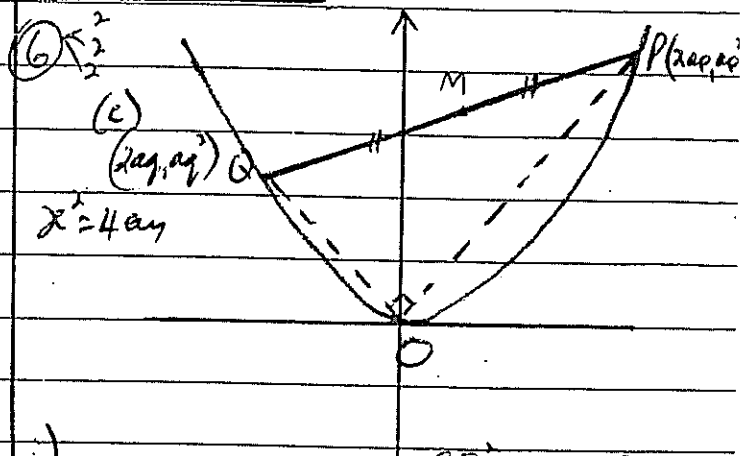
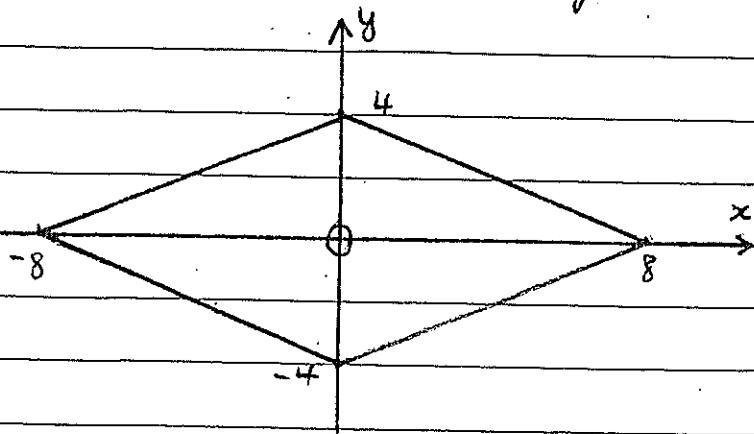
(3) (a) x, y
 $(-1, 3)$

$m_1 : m_2$
 $= 4 : -3$

Coords of reqd point
are $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$
 $= \left(\frac{4 \times 5 + -3 \times -1}{4 + -3}, \frac{4 \times -7 + -3 \times 3}{4 + -3} \right)$
 $= (23, -37)$

(6) $|x| + |2y| = 8$
(3) When $x = 0$ $|2y| = 8$
 $2y = \pm 8$
 $y = \pm 4$
When $y = 0$ $|x| = 8$
 $x = \pm 8$

Because of absolute values
 $-8 \leq x \leq 8$ and $-4 \leq y \leq 4$



i) Gradient of $OP = \frac{ap}{2ap} = \frac{p}{2}$
 $\therefore \text{Gradient of } OQ = \frac{q}{2}$

If OP is perp to OQ

$m_1 m_2 = -1$ (2)

$\frac{p}{2} \times \frac{q}{2} = -1$

$pq = -4$

ie $pq + 4 = 0$

ii) Coordinates of M , mid-point
of PQ are $\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2}$
 $= \left(a(p+q), \frac{a}{2}(p^2 + q^2) \right)$ (2)

iii) Locus of M given by

$X = a(p+q) \therefore p+q = \frac{X}{a}$

$Y = \frac{a}{2}(p^2 + q^2)$

$= \frac{a}{2} [(p+q)^2 - 2pq]$

$= \frac{a}{2} \left[\left(\frac{X}{a} \right)^2 - 2(-4) \right]$

$= \frac{a}{2} \left[\frac{X^2}{a^2} + 8 \right]$

$= \frac{X^2}{2a} + 4a$

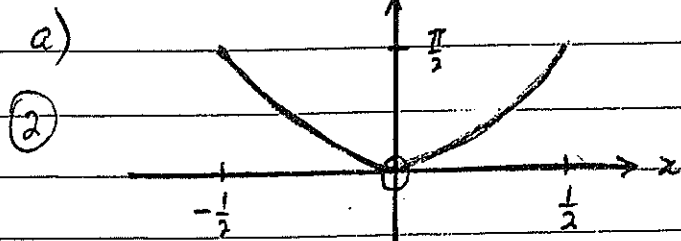
(2)

$\therefore$ Locus of M is

$2aY = X^2 + 8a^2$

$X^2 = 2aY - 8a^2$

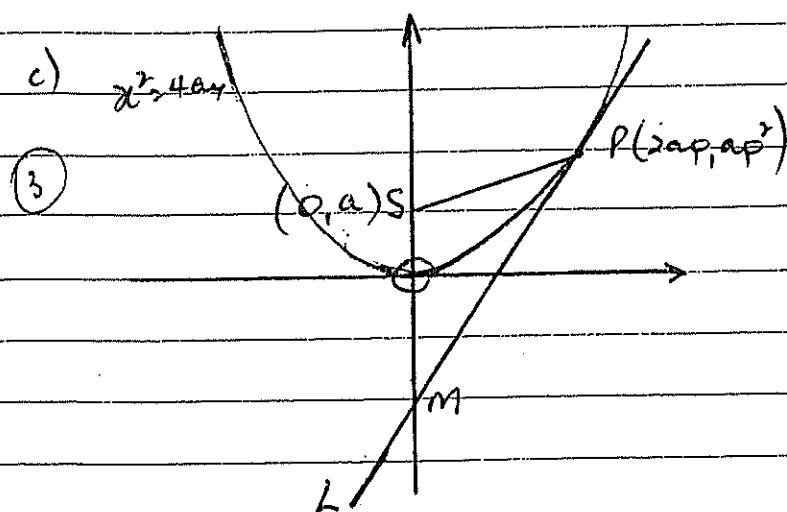
(12) Question 5:



b) i) Range of $y = \frac{-1}{2(1+x^2)}$
 (3) is $-\frac{1}{2} \leq y < 0$

ii) $\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$
 $\sin^{-1}(0) = 0$

Range of $\sin^{-1}\left[\frac{-1}{2(1+x^2)}\right]$
 is
 $-\frac{\pi}{6} \leq y < 0$



If line L meets the axis of the parabola at M , we have to prove that $\angle SPM = \angle SMP$

Proof: The tangent at P has eqn. $y = px - ap^2$
 $\therefore$ Co-ords of M (where $x=0$) are $(0, -ap^2)$
 $\therefore$ Length $SM = a + ap^2$

Length $SP = \sqrt{(ap^2 - a)^2 + (2ap - 0)^2}$
 $= \sqrt{a^2 p^4 + a^2 - 2a^2 p^2 + 4a^2 p^2}$
 $= \sqrt{a^2 p^4 + 2a^2 p^2 + a^2}$
 $= \sqrt{a^2 (p^4 + 2p^2 + 1)}$
 $= \sqrt{a^2 (p^2 + 1)^2} = a(p^2 + 1)$
 $= ap^2 + a$

$\therefore SP = SM$

$\therefore \triangle SPM$ is isosceles
 $\therefore L$ is equally inclined to SP and the axis of the parabola

(d) $S_n = 13 \times 6^n + 2 = 5M$
 (4) where M is integral

Assume S_n is $13 \times 6^k + 2 = 5M$

Prove S_{n+1} is $13 \times 6^{k+1} + 2 = 5N$
 that (where N is integral)

$$\begin{aligned} 13 \times 6^{k+1} + 2 &= 13 \times 6^k \cdot 6 + 2 \\ &= 6(13 \times 6^k) + 2 \\ &= 6(13 \times 6^k + 2) + 2 - 12 \\ &= 6(5M) - 10 \\ &= 5(6M - 2) \\ &= 5N \end{aligned}$$

where N is integral $(= 6M - 2)$

$S_1 = 13 \times 6^1 + 2 = 80$

which is a multiple of 5

Since S_1 is true then from step 1 and 2 above S_{n+1} is true $\therefore S_n$ so on for all positive integers $(n \geq 1)$

30 Marks Solution (Page 6)

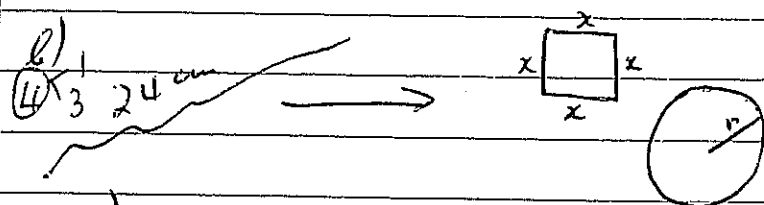
Question 6 (12)

(3) a) $\int \frac{dx}{\sqrt{1-4x^2}}$ If $x = \frac{1}{2} \sin \theta$
 $= \int \frac{dx}{\sqrt{1-4\sin^2 \theta}} \quad \frac{dx}{d\theta} = \frac{1}{2} \cos \theta$
 $= \int \frac{1}{\sqrt{1-\sin^2 \theta}} \cdot \frac{1}{2} \cos \theta d\theta$
 $= \int \frac{1}{\cos \theta} \cdot \frac{1}{2} \cos \theta d\theta$

$= \int \frac{1}{2} d\theta = \frac{1}{2} \theta + C$

If $x = \frac{1}{2} \sin \theta$ $2x = \sin \theta$
 $\theta = \sin^{-1} 2x$

$\therefore$ Integral = $\frac{1}{2} \sin^{-1} 2x + C$



i) $4x + 2\pi r = 24$

$2x + \pi r = 12$

$r = \frac{12-2x}{\pi}$

Sum of areas

$A = x^2 + \pi r^2$

$= x^2 + \pi \left(\frac{12-2x}{\pi} \right)^2$

$= x^2 + \frac{1}{\pi} (12-2x)^2$

ii) A is a maximum when

$\frac{dA}{dx} = 0$ and $\frac{d^2A}{dx^2} < 0$

$\frac{dA}{dx} = 2x + \frac{1}{\pi} 2(12-2x)(-2)$

$= 2x - \frac{4}{\pi} (12-2x)$

$= 0$

when $2x = \frac{4}{\pi} (12-2x)$

$2\pi x = 48 - 8x$

$(2\pi+8)x = 48$

$x = \frac{48}{8+2\pi} = \frac{24}{4+\pi} \approx 3.36 \text{ cm}$

$\frac{d^2A}{dx^2} = 2 - \frac{4}{\pi} (-2) = 2 + \frac{8}{\pi}$

> 0

$\therefore$ This turning point gives minimum area ($\approx 20.16 \text{ cm}^2$)

$\therefore$ Need to test the end points
 at $x = 0$ and $x = 6$

When $x = 0$, radius of circle = $\frac{12}{\pi}$
 and Area = $\pi \left(\frac{12}{\pi} \right)^2 = 45.8 \text{ cm}^2$

When $x = 6$, area of square = 36 cm^2

$\therefore$ Maximum area occurs when $x = 0$

ie when

$A = 0 + \frac{1}{\pi} (12)^2 = \frac{144}{\pi} \text{ cm}^2$

c) $\frac{1}{|3-x|} \geq \frac{1}{5}$

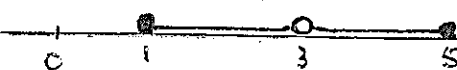
$|3-x| \leq 2, 3-x \neq 0$

$-2 \leq 3-x \leq 2, x \neq 3$

$-5 \leq -x \leq -1$

$5 \geq x \geq 1$

ie $1 \leq x \leq 5$, excluding $x = 3$

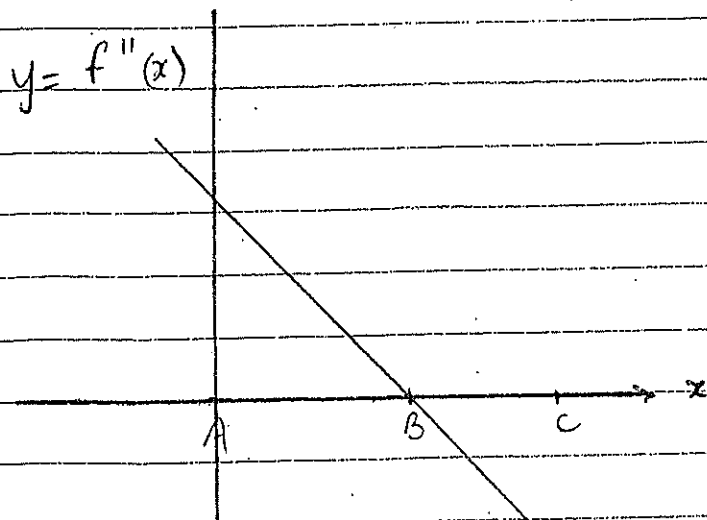
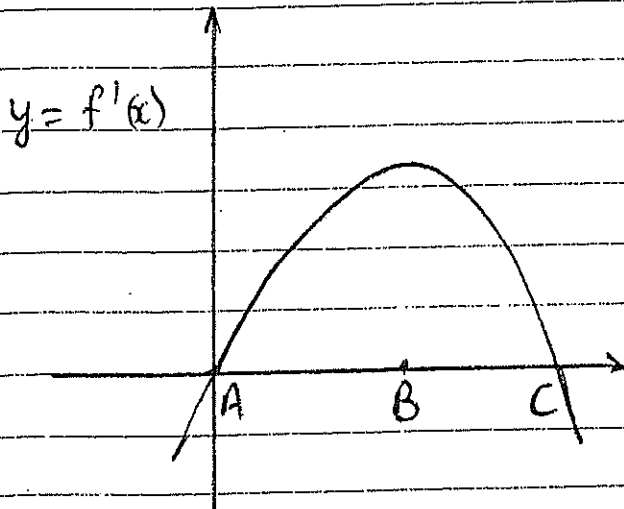
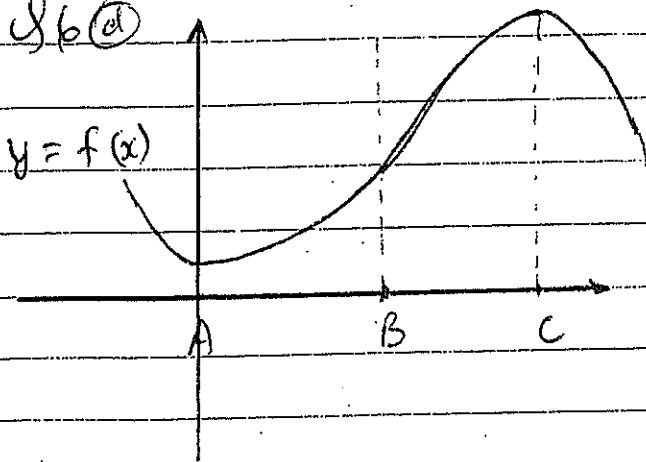


③ $\leftarrow \frac{2}{0}$

86④

3U Maths Solutions (Page 7)

Notes Min t. point at A
Pt of inflexion at B
Max t. pt at C



Notes: +ve to left of B
0 at B
-ve to right of B

[Not necessarily
a straight line]

Question 7. (12)

3) a) $x^2 + mx + n = 0$
with roots α, β

i) $\alpha + \beta = -m$

ii) $\alpha\beta = n$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= m^2 - 2n$

iv) $3\alpha^2\beta^2 + 3\alpha\beta^2$
 $= 3\alpha\beta(\alpha + \beta)$
 $= 3n(-m) = -3nm$

b) i) $\int_{-1}^1 x^2 \cos x \, dx$
3) $= 2 \int_0^1 x^2 \cos x \, dx$?

Consider the graph of the function $y = x^2 \cos x$ with Key points $(-1, \cos 1)$
When $x = -1$

$$y = (-1)^2 \cos(-1)$$

$$= 1 \cos 1$$

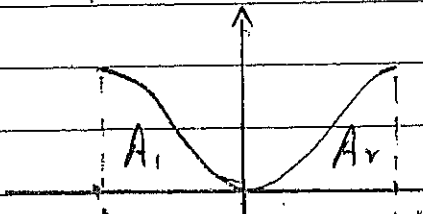
as $\cos(-\theta) = \cos \theta$

When $x = 1$, $y = \cos 1$

When $x = 0$, $y = 0$

$\therefore$ As $f(x) = f(-x)$

The function is even
and its graph is
symmetrical about
the y axis



$\therefore A_1 = A_2$

$\therefore$ This statement is TRUE

ii) $\int_{-3}^3 (x \cos x + \sin x) \, dx = 0$?

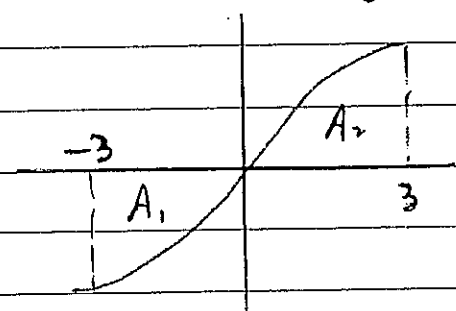
$f(-3) = -3 \cos(-3) + \sin(-3)$
 $= -3 \cos 3 - \sin 3$

$f(3) = 3 \cos 3 + \sin 3$

This function is odd

i.e. $f(x) = -f(-x)$

so has point symmetry
about the origin



so $A_1 = -A_2$

$\therefore A_1 + A_2 = 0$

$\therefore$ This statement is TRUE

[Note: It is sufficient to
show that i) graph is even
and ii) is odd. The
actual shape of each
curve is irrelevant]

3U Maths Solns (Page 9)

3) 87c) $\frac{d}{dx} (2x \tan^{-1} x - \ln(1+x^2))$

$$= \tan^{-1} x \cdot 2 + 2x \cdot \frac{1}{1+x^2}$$

$$= \frac{2}{1+x^2} \cdot 2x$$

$$= 2 \tan^{-1} x$$

Hence $\int \tan^{-1} x \, dx$

$$= \left[\frac{1}{2} (2x \tan^{-1} x - \ln(1+x^2)) \right]_0^1$$

$$= \frac{1}{2} (2 \tan^{-1} 1 - \ln 2)$$

$$= \frac{1}{2} (0 - \ln 1)$$

$$= \tan^{-1} 1 - \frac{1}{2} \ln 2$$

$$= \frac{\pi}{4} - \frac{\ln 2}{2}$$

3) d) $x^2 + y^2 - 2x - 14y + 25 = 0$

i) If the line $y = mx$ meets this circle, then

$$x^2 + (mx)^2 - 2x - 14 \cdot mx + 25 = 0$$

$$(1+m^2)x^2 - (2+14m)x + 25 = 0$$

If the line is to intersect the circle then this equation will have two real different roots

$$\Delta > 0$$

$$b^2 - 4ac > 0$$

$$(-2(1+7m))^2 - 4(1+m^2) \cdot 25 > 0$$

$$4(1+7m)^2 - 100(1+m^2) > 0$$

$$(1+7m)^2 - 25(1+m^2) > 0$$

ii) If $y = mx$ is a tangent to the circle, then the above quadratic eqn in x will have equal roots i.e. $\Delta = 0$

$$(1+7m)^2 - 25(1+m^2) = 0$$

$$1 + 14m + 49m^2 - 25 - 25m^2 = 0$$

$$24m^2 + 14m - 24 = 0$$

$$12m^2 + 7m - 12 = 0$$

$$(4m-3)(3m+4) = 0$$

$$m = \frac{3}{4} \text{ or } -\frac{4}{3}$$

NOTE: The following questions were taken from past HSC papers

All Question 1

All Question 2

Question 5 (d)

Question 7 (d)